

The Mental Models of Leonardo da Vinci: Applications to Real-World Contexts

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Abstract

In the broader context of the theme of "lines of force," we can see how Leonardo's mental models can indicate a uniquely polymathic cognitive architecture, with profound implications for how we might tackle contemporary complex problems, including the design of future AI systems. This paper distills ten mental models from Leonardo da Vinci's manuscripts, drawings, and paintings — principles that governed how he observed, reasoned, and created across domains we have since forced apart. Drawing upon decades of scholarly engagement with his corpus, we examine these models not as historical curiosities but as living frameworks: for education, for environmental stewardship, for scientific practice, and for the recovery of an integrative intelligence that specialisation has progressively eroded. Each model is grounded in specific works and manuscripts, and each is illustrated by a contemporary exemplification — from molecular geometry and ensemble weather forecasting to biomimetic engineering and planetary governance — illustrating that Leonardo's cognitive architecture continues to generate breakthrough outcomes across diverse fields. Where the models operate individually, they illuminate; where they operate together, as three synthesis examples drawn from medicine, genomics, and international environmental law suggest, they produce the conditions under which problems of genuine complexity yield. Leonardo's refusal to separate art from science, body from earth, theory from experience, or the known from the unknowable constitutes, we argue, a coherent cognitive architecture whose relevance has only deepened as the consequences of fragmented thinking become ever more visible.

Introduction

This paper extracts principles from Leonardo da Vinci's practice to address contemporary challenges in design, engineering, scientific research, product innovation and sustainable policy. After decades of scholarly engagement with Leonardo's work, the question that emerges is not merely historical: Why does Leonardo matter today? What fundamental insights can we derive from his practice regarding the propensity to innovate and the approach to tackling complex problems? And how might these inform the intelligent systems we design in the future?

This reflection employs a series of well-known images and concepts from Leonardo's oeuvre to extract aphoristic principles—some philosophical, others with direct implications for breakthrough thinking in various fields. Whilst this approach departs from strictly descriptive historical analysis, it serves to illuminate the profound relevance of Leonardo's integrative thinking to our present moment.

Leonardo's case is particularly interesting in that it serves as an archetype for polymathic thinking - defined here as exceptional cross-domain versatility combined with a level of

multi-perspectival integration that generates novel insight (Ahmed, 2019). His breakthrough innovations, often centuries ahead of their time, were achieved through what we might today characterise as interdisciplinary, transdisciplinary, and systems thinking.

Several important accounts of Leonardo's epistemology and cognitive practice have preceded this work and inform its approach. Kemp's foundational studies of Leonardo's science of art (1981; 2006) establish the structural relationship between visual investigation and theoretical inference that underlies several of the models proposed here. Pedretti's editorial work on the manuscripts (1977) and Bambach's catalogue of the drawings (2003) provide the philological grounding without which claims about Leonardo's cognitive patterns cannot be reliably made. Fehrenbach's analysis of Leonardo's fluid dynamics experiments (1997) is directly relevant to Models IV and VIII, while Laurenza's account of anatomical method (2012) bears on Models II and VI. What distinguishes the present paper from these contributions is not a new claim about what Leonardo did, but a systematic attempt to formalise the cognitive principles implicit in his documented practice into transferable frameworks — 'mental models' in the sense developed below — and to test their contemporary generativity against documented cases of innovation across multiple fields.

What is a mental model?

The term 'mental model,' as used in this paper, requires a brief definition before the models themselves are introduced. In cognitive science, the concept originates primarily with Johnson-Laird (1983), who proposed that reasoning proceeds not through formal logical operations but through the construction and manipulation of internal representations that are structural analogues of the states of affairs they represent — small-scale working simulations that can be run forward to generate predictions and revised when those predictions fail against experience. This is distinct from a heuristic (a decision rule that bypasses systematic reasoning) and from a schema (a categorical structure organising prior knowledge). We adopt the term here in a sense broadly compatible with Johnson-Laird's, though applied not to individual cognitive acts but to stable, recurring patterns in Leonardo's investigative practice — patterns that operated across domains, generated testable predictions, and were revised through the theory-experience dialectic described in Model V. The ten models are not exhaustive of Leonardo's cognitive repertoire, nor are they mutually exclusive: they are, as the synthesis section argues, aspects of a single integrative mode of mind, isolated here for analytical clarity.

A Note on the Method

The ten models presented here were derived through systematic thematic analysis applied to Leonardo's principal manuscript sources — the Codex Atlanticus, the Codex Leicester, Manuscript E, the Windsor anatomical folios, and the Paris manuscripts A through M — in the standard critical editions (Pedretti, 1977; Reti, 1974; Kemp and Roberts, 1989). Analysis proceeded in two stages. In the first stage, recurring investigative patterns were identified inductively across the corpus: moments at which Leonardo explicitly compares phenomena from different domains, revises a theoretical position in response to experimental evidence, acknowledges the limits of observational certainty, or draws on mathematical structure as the

primary explanatory framework. In the second stage, these patterns were grouped into six overarching themes: the unity of nature, working with nature, seeing and uncertainty, mathematics of everything, the art of science, and lessons from Leonardo. This latter theme became an interpretive attempt to relate Leonardo's thinking as timeless principles with implications for contemporary challenges. These were shortlisted to a set of ten, which was tested for internal coherence and for external generativity against documented cases of innovation outside Leonardo's own practice. Two of the authors reviewed candidate models independently before the final selection was agreed.

Regarding the selection of contemporary examples that illustrate the models, these were researched and measured against a fourfold criteria: the complexity of the problem, the relevance of the approach to one or more mental models, the number of Leonardo mental models exemplified, and the magnitude of the breakthrough (as per the impact of the outcome). Some of these examples have directly cited Leonardo as an influence, whereas others have not, even if the correspondence with one or more of his mental models is strongly evident.

I. Geometry and the Mathematical Structure of Reality

The *Vitruvian Man* exemplifies Leonardo's fundamental conviction that geometrical proportions and arithmetical ratios govern the visible and invisible world. Among the mathematical sciences, geometry holds a privileged position. Where arithmetic and algebra offer symbolic representations, geometry represents spatial relationships that resist pure symbolic expression, enabling reasoning about multi-dimensional structures. This preference reflects not merely personal taste but a fundamental epistemological commitment to the further reaches of mathematical knowledge.

Geometry's power lies in its capacity to articulate relationships that resist purely arithmetic expression. The ratio of a circle's radius to its circumference, represented by π , exemplifies this irrational dimension. This capacity for geometric reasoning allowed Leonardo to intuit structural forms that would only later be formalised mathematically - a practice of discovery as much as representation.

This commitment is demonstrated in Leonardo's geometrical knot patterns for the *Accademia Leonardi*, which treat geometry as a form of divine beauty, and in the dodecahedron illustrations for Luca Pacioli's *De Divina Proportione*, where the skeletal "vacuous forms" demonstrate geometric transformation. In both cases, formal structure and aesthetic perception are inseparable - the mathematical pattern is also the beautiful pattern.

This principle has directly inspired Vebjørn Sand, whose realisation on encountering Leonardo's 1502 Golden Horn bridge sketch — that a single parabolic arch, requiring no fasteners, mortar, or lateral support, could resolve all compressive forces through geometry alone — led to the construction in 2001 of the first of Leonardo's civil engineering designs ever built at full scale, confirmed structurally sound by MIT researchers in 2019 using a 1:500 blockwork model replicating fifteenth-century material constraints (Sand, 2001;

Deseret News, 2001). The bridge demonstrates that for Leonardo mathematical structure was not an abstraction applied to physical problems but the physical reality itself: the arch does not merely describe the distribution of forces, it is that distribution, made visible and inhabitable.

The representation of such a mental model can be seen as recently as 2026. The TongGeometry framework demonstrates that the three-dimensional geometric structure of molecules — the precise spatial arrangement of atoms and the angular relationships between chemical bonds — constitutes the primary information required to predict biological activity, and that models which reason about geometry rather than symbolic sequences achieve markedly superior results in drug discovery and materials design (Zhang et al., 2026). Where conventional approaches encoded molecules as graphs of atoms and bonds, TongGeometry treats the molecule as a geometric object in three-dimensional space, recovering the conviction that animated Leonardo's preference for geometry over arithmetic: that the thing itself, encountered in its spatial fullness, discloses structural truths that abstraction conceals.

II. The Unity of Human and Natural Systems

Leonardo's anatomical studies reveal a crucial insight: the human body is constituted from the same material components as the Earth itself. This principle extends the geometric foundation of Model I - if natural and human systems share mathematical structures, geometric analysis reveals these isomorphisms. Leonardo's cross-domain investigations systematically sought such structural parallels.

The branching patterns of bronchial passages mirror those of trees and river deltas. These are not mere analogies but reflect universal optimisation principles: branching patterns maximise surface area whilst minimising material, whether in biological or geological systems. Modern graph theory formalises these isomorphisms through topological invariants; Leonardo identified them through sustained visual investigation.

The concept of the "Minor World" - in which the human body is a microcosm of the macrocosm—was a common Renaissance philosophy, but Leonardo applied it with unprecedented scientific rigour. His views on the Earth as a living organism are detailed primarily in the *Codex Leicester*, where he describes the planet as dynamic rather than static: soil as flesh, rocks as bones, water as blood. He challenged the biblical view of geological time, recognising that fossils on mountains meant those areas were once underwater—evidence of massive, slow change. Water, in his view, was the primary agent of transformation, shaping landscape through erosion and sediment transport.

Leonardo's general refusal to separate phenomena we would now classify differently is exemplified by the "theme sheet" containing geometry, drapery, trees, hair, and water simultaneously, and in his note explicitly comparing hair movement and water movement: they follow, he states, "the same rule."

This principle was employed In the late 1980s, when the Japanese engineer Eiji Nakatsu was tasked with redesigning the nose of the Series 500 Shinkansen to eliminate the sonic boom produced when the train emerged from tunnels at speeds exceeding 300 kilometres per hour. Nakatsu, an amateur birdwatcher, identified the operative structural isomorphism: the kingfisher (*Alcedo atthis*) transitions between two media of drastically different densities — air and water — without perceptible disturbance, because its bill is geometrically optimised to manage the pressure differential at the interface. The tunnel problem, whilst different in specifics, is structurally identical: the train moves between two pressure environments and must manage the shock at the transition without producing a wave. Modelling the nose on the cross-sectional profile of the kingfisher's bill reduced tunnel boom by thirty per cent and decreased energy consumption by fifteen per cent whilst increasing speed by ten per cent (Benyus, 1997). This is the cognitive operation that Leonardo performs on his theme sheet — the recognition that hair and water, bird and train, biological and technological system, obey the same underlying rule — enacted not as metaphor but as engineering solution.

There are also examples of significant breakthroughs using this model as more directly attributable to Leonardo. Working from Leonardo's scattered notebook pages on the robotic knight, Mark Rosheim reconstructed the geometric logic of Leonardo's joint designs — nested pulleys, cables, and gears arranged to reproduce the angular ranges of the human shoulder and wrist — and used those principles to design NASA's Surrogate humanoid robot, demonstrating that the boundary Leonardo refused to draw between the study of living bodies and the construction of machines was not a Renaissance idiosyncrasy but a structural insight with direct engineering consequences (Rosheim, 1998; Rosheim, 2006). Leonardo's robotic knight was not an imitation of the human body but an investigation of it: the machine was the instrument of anatomical understanding, and the anatomy was the specification of the machine — a unity that Rosheim's work makes technically precise rather than metaphorically suggestive.

III. The Earth as Living Body

If the Earth functions as a complex, self-regulating system—what Leonardo termed *vene d'acqua* (veins of water)—then understanding its behaviour becomes both a scientific and an ethical responsibility. This conception, resonant with contemporary formulations such as the Gaia hypothesis (Lovelock, 1979), suggests that the planet deserves the same consideration we accord to any living body. The hubris of presumed human dominion over nature has been repeatedly challenged by geological and political events that reveal the limits of such presumptions.

The behaviour of terrestrial elements follows mathematical rules; understanding these constitutes a primary human responsibility. During our brief existence on Earth, the fundamental task involves comprehensive understanding—both emotional and intellectual—of planetary operations and celestial mechanics. This defines what education should ultimately aim to achieve: not the compartmentalised mastery of discrete subjects, but holistic understanding of natural systems in their interconnection.

Today, the model of Earth as a living body has most aptly been brought to life by the Planetary Boundaries framework, proposed by Johan Rockström and colleagues in *Nature* in 2009. It identifies nine Earth-system processes — including climate change, biosphere integrity, freshwater use, and novel chemical entities — within which human activity must remain if the planet is to maintain the stable conditions in which human civilisations have developed (Rockström et al., 2009). The framework does not describe the Earth as a collection of separate environmental problems but as a single, coupled, self-regulating system with its own thresholds of tolerance, beyond which the capacity for self-correction is compromised in ways that cannot be reversed on human timescales. The framework translates the mental model into a quantitative instrument for contemporary policy, without diminishing the organic, systemic conception of the planet from which it proceeds.

Ray Anderson's 1994 epiphany — reading Paul Hawken's account of industrial ecology and recognising that Interface's take-make-waste model was treating the Earth not as a self-regulating living body with its own thresholds of tolerance but as an inexhaustible resource — led to a restructuring of an entire industrial enterprise around the principle (Anderson, 2009; Hawken, 1993). The Factory as a Forest programme, developed with Biomimicry 3.8, makes this principle operational at the level of manufacturing design, while the ninety-six per cent reduction in greenhouse gas emissions achieved between 1996 and 2019 demonstrates that Model III is not a poetic aspiration but a design constraint with measurable commercial consequences (Shourkaei et al., 2024).

It should be noted that Leonardo's geological account was not uniformly correct: his hydraulic theory of landscape formation, while prescient in its dynamism, underestimated the role of tectonic processes and contained errors in his understanding of river sedimentation rates that were not corrected until the nineteenth century (Kemp, 1981, pp. 214–217). The value of the model lies not in the accuracy of every specific claim but in the investigative disposition it instantiated.

IV. Chaos, Predictability, and Natural Systems

The behaviour of water exemplifies complex dynamics within chaotic systems and raises fundamental questions about the limits of predictability. A critical distinction must be maintained: chaotic systems are not random, nor are they deterministic in the classical sense. They operate according to probabilistic rules within parametric constraints - which is why weather prediction improves incrementally without ever achieving complete accuracy.

This distinction carries direct computational implications: deterministic models fail catastrophically on chaotic systems, whilst probabilistic models waste resources on deterministic problems. Any system aspiring to Leonardian reasoning must first classify the nature of a problem domain before selecting an appropriate solution strategy. Recognising which phenomena resist precise forecasting prevents the misapplication of deterministic models to inherently probabilistic systems—a failure mode with serious consequences in both scientific practice and policy formation.

Leonardo's experimental tank drawings from the *Codex Leicester*—showing water entering through a *bocca* (mouth), with instructions to use millet seed and sand to track wave behaviour—demonstrate that he was not merely observing fluid dynamics but systematically probing the boundary between the predictable and the indeterminate. His experimental method anticipated modern understandings of chaotic systems (Lorenz, 1963), treating unpredictability not as a failure of knowledge but as a property of the phenomenon itself.

The European Centre for Medium-Range Weather Forecasts introduced operational ensemble forecasting in 1992, founding the technique on a direct application of chaos theory to the practice of meteorology: because the atmosphere is a chaotic system exquisitely sensitive to initial conditions — as Lorenz had demonstrated theoretically in 1963 — no single deterministic forecast can reliably represent what will happen, and the only epistemically honest approach is to run many forecasts from slightly different initial states and present the range of outcomes as a probability distribution (Molteni et al., 1996). The ECMWF's operational transition from deterministic to ensemble forecasting is, in methodological terms, the adoption of exactly the stance Leonardo arrives at through observation of turbulent water — treating unpredictability not as a failure of knowledge but as a property of the phenomenon, to be quantified and communicated rather than suppressed or disguised.

V. The Dialectic of Theory and Experience

Deep understanding requires constant interchange between experiential knowledge and theoretical abstraction. Computationally, this dialectic operates as iterative refinement: theoretical models generate predictions, experimental validation reveals discrepancies, models update accordingly. The cycle continues until theory and observation converge within acceptable error bounds. This differs fundamentally from pure empiricism (data without theory) and pure rationalism (theory without testing). Systems that privilege theory over observation generate elegant but wrong solutions. Systems that privilege observation over theory cannot generalise beyond their training data. Leonardo's dialectic avoids both failure modes.

Leonardo's pyramid of force exemplifies abstract conceptualisation applicable to diverse natural forces—gravity, spring mechanics, projectile motion. Yet such conceptions must continuously align with experiential verification. The historical reluctance of eighteenth- and nineteenth-century chemists to prioritise theory impoverished their discipline until Mendeleev's periodic table demonstrated the power of systematic theoretical frameworks.

Leonardo's water pressure experiment with the leather-sided tank and weighted planks demonstrates quantified experiential testing of an abstract hydraulic principle. And his glass model of the aortic valve—later verified by Caltech engineer Morteza Gharib using modern imaging techniques (Gharib et al., 2002) - stands as perhaps the most striking example of theory tested through physical experimentation across centuries. We must respect both the generalisers who produce mathematical insights and the empiricists who test these against the resistance of physical reality. This grit complicates overly systematised theories, a principle

applicable not only to natural philosophy but to institutional and political understanding as well.

When Ignaz Semmelweis introduced mandatory handwashing at the Vienna General Hospital in May 1847, maternal mortality fell from approximately sixteen per cent to below two per cent within months — one of the most dramatic interventions in the history of medicine (Semmelweis, 1861). And yet his discovery was dismissed and his career effectively destroyed, because he could offer no theoretical explanation for what the empirical data showed: the germ theory required Pasteur’s experimental work in the 1860s to supply the conceptual framework that made Semmelweis’s evidence comprehensible, and only then did the intervention gain the acceptance that the data alone had failed to secure. This sequence enacts, with particular force, the failure mode that Leonardo’s dialectic of theory and experience is designed to prevent: the condition in which experience is generating decisive knowledge that the available theoretical framework is not equipped to receive, so that the community’s instruments of recognition cannot see what the evidence is showing. Semmelweis observed the edge of something significant; the community lacked the theoretical apparatus to perceive what the observation disclosed.

VI. Structural Intuitions: Art and Science

The design of organic forms and processes obeys mathematical rules, yet understanding them requires what might be termed “structural intuitions”—moments of recognition in which an observer becomes aware that something significant is occurring in the phenomenon before them (Kemp, 2016). Scientists respond by developing formal models with testable predictions; experimental validation constrains the model space until few viable theories remain. Artists present their understanding through open-ended representations that invite multiple interpretations. Both modes originate in shared intuitive recognition; their divergence lies in methodology and presentation.

This model presents one of the hardest challenges for artificial intelligence: multi-objective optimisation with non-decomposable objectives. Standard approaches optimise aesthetic and functional criteria separately and then combine results. Leonardo’s method demands simultaneous satisfaction: the aesthetic is the functional. A bridge’s beauty emerges from optimal force distribution. A painting’s composition reflects optical principles of visual attention. Solutions that excel in one dimension whilst failing another are rejected entirely. Formalising structural intuition computationally requires neural-symbolic integration—neural pattern recognition (intuition) operating alongside symbolic verification of principles (analysis). Current AI architectures largely lack this dual-processing capability.

Leonardo demonstrates structural intuition throughout his practice. In the *Star of Bethlehem* botanical drawing, he remakes rather than merely records the specimen, creating an “ideal” form that captures the rule behind the appearance. In the *Ginevra de’ Benci* portrait, the curling of hair intuitively follows water dynamics before Leonardo had theorised that connection explicitly. Cardiac surgeon Francis Wells has confirmed that Leonardo’s ox heart drawings and dissection methods remain in use in surgical teaching today (Wells, 2013). To

separate artistic insight from analytical scientific rigour impoverishes both our perceptual and our cognitive capacities. Their reintegration remains an urgent task.

The Alamillo Bridge in Seville, completed by Santiago Calatrava for the 1992 World Exposition, is constituted by a single inclined concrete pylon and thirteen pairs of stays connecting pylon to deck: a structure whose visual drama is entirely explained, and entirely generated, by the structural logic of its load distribution (Casas and Aparicio, 1998). There is no ornament applied to the structure; the aesthetic is the structural solution, simultaneously and inseparably. Calatrava, trained as both an architect and a civil engineer, approaches structural design through the same cognitive sequence that Leonardo describes and practises: the structural intuition precedes and generates the formal solution, so that the rightness of the form is perceptible before the calculations confirm it. This is the non-decomposable objective that represents one of the hardest challenges for any design methodology that separates aesthetic and engineering criteria: the Alamillo exists precisely because those criteria were never separated, and because the perception of structural form — the recognition of the rule behind the shape — was the operative faculty throughout.

The naming of Intuitive Surgical's robotic system — prototype "Lenny" (Leonardo), then "Mona" (Mona Lisa), then "da Vinci" — was a deliberate institutional acknowledgement that the engineering challenge at the heart of minimally invasive surgery is precisely the challenge Leonardo had identified in his anatomical notebooks: that the human body's mechanical logic, understood as geometry rather than merely as anatomy, can be reproduced in a machine that moves with the precision and range of the structures it replaces (PMC, 2018; 2023). The da Vinci system, now used in over six million procedures worldwide, embodies the structural intuition that art and science are a single act of perception — that the surgeon's console and the robotic wrist are the descendants not of two separate traditions but of one integrative intelligence.

Steve Jobs cited Leonardo explicitly and repeatedly as the model for the conviction that drove Apple's product philosophy: that technology and the liberal arts are not two domains trading off against each other but a single problem whose solution requires the thinker and the maker to be the same person, as Leonardo — who mixed his own pigments, ground his own lenses, and dissected the cadavers he painted — embodied (Isaacson, 2011). The product that resulted from this structural intuition — the Macintosh, the iPod, the iPhone — enacted the same perception Leonardo had made the foundation of his practice: that the deepest forms of beauty are not decorative additions to functional objects but the visible expression of the object's inner structural logic.

VII. Scale, Power, and Respect for Natural Force

Forces that sustain life at human scales can prove destructive at larger magnitudes. The development of large-scale machinery, analytical capabilities, and powerful technologies amplifies human impact exponentially - yet our fundamental fragility, and the planet's

vulnerability, remains constant. Failure to respect natural force and act in harmony with natural systems invites catastrophic consequences.

Water exemplifies this principle across scales. Properly understood and employed—as demonstrated by Archimedean screws and the full range of hydraulic engineering schemes, ancient and modern—water provides immense benefit. Yet its power can prove terrifying, as tsunamis demonstrate. The cosmic-scale energy contained in wave and wind far exceeds current generation capacity, but approaching it requires appropriate humility and understanding of the force being harnessed. In Leonardo's deluge drawings, battle tents (*padiglioni*) are torn into the sky—a vivid image of nature's indifference to human presumption.

Leonardo's reference to the *Scala della Sforzesca* di Vigevano—where water was stepped down a hillside to prevent erosion, a structure still surviving today—presents a model of working productively with natural force at scale rather than attempting to dominate or redirect it.

The Three Gorges Dam on the Yangtze River, completed in 2006, generates approximately one hundred terawatt-hours of electricity annually and was designed to provide flood control, improve navigation, and reduce dependence on coal. It has achieved all three purposes. It has also triggered mass upstream sedimentation, disrupted the migratory patterns of the Yangtze river dolphin and Chinese sturgeon, altered downstream hydrology in ways affecting both riverine ecosystem and local climate, induced geological instability in the reservoir zone, and required the displacement of approximately 1.3 million people (Wu et al., 2012). The example is presented here not as a failure but as an enactment, at extraordinary scale, of the precise relationship between scale and consequence that Leonardo identifies: forces manageable and beneficial at human scale become destructive when amplified, and the planet's tolerance for disturbance does not grow proportionally with the ambition of the engineering. The Yangtze's force was not destroyed by the dam; it was redirected, delayed, and concentrated in forms that the original design did not anticipate and that the system is still absorbing. Leonardo's deluge drawings are not fantasies; they are predictions.

VIII. Working with Nature Rather Than Against It

The principle that nothing comes from nothing—*ex nihilo nihil fit*—is fundamental. Leonardo's investigations of perpetual motion, including observations of Venetian experiments, reinforced this ancient truth. Contemporary discourse must resist promises of something from nothing; thermodynamic limits are not negotiable.

Water serves as a subtle ally across scales, yet it cannot be forced against its nature. The Florentine attempt to divert the Arno past Pisa through artificial channels illustrates the cost of misunderstanding this principle. Leonardo learnt that water, when properly understood, enables human vision; when coerced, it ultimately defeats human intentions. This lesson extends beyond hydraulic engineering to broader environmental relationships and to any

design process that attempts to override rather than collaborate with the properties of the materials and systems involved.

Leonardo's Archimedean screw drawings are an early exploration of harnessing rather than fighting water's natural behaviour. His Arno canal proposal via Prato and Pistoia—following exactly the route of the modern motorway—showed Leonardo working with the landscape's natural logic rather than against it.

Such an approach to sustainable engineering can be seen in The Dutch Room for the River programme, authorised in 2006 and completed by 2019, addressed the increasing flood risk to four million people in the Rhine–Maas–Waal delta not by raising dykes further — the traditional response — but by lowering floodplains, relocating polders, removing obstacles from riverbeds, and creating side channels that allow the river to spread into designed overflow areas, reducing peak water level and the force concentrated against existing flood defences (Ministry of Infrastructure and the Environment, 2014). The key conceptual shift is from resistance to accommodation: rather than forcing the river to remain within an ever-narrower channel under ever-greater hydraulic pressure, the programme gives the water room to move in ways that serve human safety whilst working with the river's own logic. The programme also won the Rosa Barba Prize for landscape architecture, a recognition that the functional and the aesthetic were, as at the Alamillo Bridge, the same thing.

IX. Vision, Perception, and the Limits of Knowledge

Understanding visual perception requires recognition of fundamental constraints. Light travels at finite speed; what we perceive occurred in the past, however recent. Our sensory apparatus provides a limited, actively constructed experience even of the immediately observable world. To understand perception, we must appreciate both the capabilities and the limitations of the eye-brain system.

Leonardo's eye anatomy drawings, which use the sectioned onion as an analogy for layered structure, represent a novel graphic method for demonstrating stratified complexity. His wax ventricle injection drawings show the transition from theoretical models of vision to direct physical investigation. The *Lumen Cinereum* analysis in the *Codex Leicester* demonstrates simultaneous contrast and the eye's active role in distorting what we perceive.

The neuroscientific investigation of perception addresses extraordinarily complex questions that remain active areas of research; the totality of cognitive processes involved continue to be revealed. This recognition should instill appropriate humility regarding the scope of human understanding. Leonardo's statement that "the eye does not know the edge of any body," from Manuscript E (c. 1508), marks his late loss of faith in optical certainty—a recognition that perception is constructive, not transparent.

This principle can be seen in the predictive coding framework in neuroscience, developed theoretically by Rao and Ballard in 1999 and extended by Karl Friston into the broader free-energy principle of brain function, proposes that the visual system does not passively receive and process sensory input but actively generates predictions about incoming signals,

comparing these predictions against what actually arrives and updating the internal model only in response to discrepancies (Rao and Ballard, 1999; Friston, 2010). What we perceive, on this account, is largely what the brain expected to perceive; the deviation between prediction and reality is both the signal of something genuinely new in the environment and the mechanism by which the model updates. The history of science is in large part a history of failed predictions, misread instruments, and anomalies dismissed as error — events whose explanation requires not better data but a better prior model.

X. The Unknown Unknowns

Beneath accumulated knowledge lies a more troubling category: phenomena whose very existence we cannot yet formulate as questions. Beyond the acknowledged gaps in our understanding of known phenomena lies what we might call the domain of unknown unknowns—a concept ancient in philosophy, though Donald Rumsfeld gave it contemporary currency. Contemporary physics of space-time confronts this challenge acutely. In cosmology and fundamental physics, we encounter potential dimensions of reality that may exceed our cognitive grasp. Has the human intellect—despite its remarkable achievements, including the comprehension of multi-dimensional geometry—reached or approached its inherent limits?

Within science itself, Heisenberg’s uncertainty principle (Heisenberg, 1927) and Gödel’s incompleteness theorems (Gödel, 1931) establish that certain limits are not merely practical but structural. String theory and related frameworks may resist empirical verification in traditional experimental terms. When our investigative equipment participates in the phenomena under examination, objectivity becomes genuinely problematic. These questions, whilst particularly pressing in advanced theoretical physics, have haunted epistemology throughout intellectual history.

For Leonardo, this recognition became most apparent in his later works. In the *Virgin, Child and St Anne* cartoon, the knowing smile of St Anne signals an awareness of secrets that lie beyond rational articulation. The late deluge drawings—combined with his written accounts—are where physical description and emotional expression merge at the threshold of what can be said. The *Salvator Mundi*, with its deliberately blurred features and rock crystal sphere, represents the boundary between the knowable (the sphere of fixed stars) and the unknowable (what lies beyond). In *St John the Baptist*, Leonardo’s last painting, optical uncertainty becomes theological: the finger points toward what cannot be seen or known.

When the James Webb Space Telescope published its first deep field results in 2022, one early finding was so unexpected that the research team initially suspected instrumental artefact: the observation of a population of massive, already-mature galaxies at redshifts corresponding to approximately 600 million years after the Big Bang — a period in which standard cosmological models predicted galaxy formation was still in its earliest stages (Labbé et al., 2023). The finding did not merely challenge an existing theory; it challenged the framework within which theories of early galaxy formation had been constructed. The JWST was not built to disprove standard cosmology; it was built to see further into what was believed to be a known domain. What it revealed instead was an unknown unknown: not a

question that had been asked and left unanswered, but a configuration of reality that the available conceptual framework had not equipped anyone to look for. This is the category of knowledge that Leonardo identifies in his late works — the domain beyond the boundary of what inquiry has yet been able to formulate as a question — and it is the category that the most consequential scientific instruments consistently disclose, not because they are designed to do so, but because genuine encounters with the natural world routinely produce results for which existing frameworks have no name.

Integrating the Models as a Unified Framework

The ten mental models set out in the preceding sections are not a taxonomy of separate intellectual virtues. They are, taken together, a description of a single mode of mind — one in which geometric precision, cross-domain structural recognition, ethical responsibility towards living systems, calibrated humility before chaos, the sustained alternation of theory and experience, the unity of aesthetic and analytical perception, respect for natural force, collaboration with rather than coercion of the systems one inhabits, honest acknowledgement of the limits of what instruments can see, and rigorous attention to what cannot yet be named all operate simultaneously and in mutual reinforcement.

Individually, each model can be demonstrated by examples of genuine but bounded achievement. What the integrated framework predicts, and what historical evidence confirms, is that breakthroughs of the deepest order — those that open new domains rather than extend existing ones — tend to arise when most or all of the models are active together. This is not an argument for intellectual dilettantism. It is an argument about the conditions under which problems of genuine complexity yield.

Three contemporary scientific breakthroughs illustrate what this synthesis looks like in practice. Karikó and Weissman's mRNA vaccine platform — twenty-five years of rejected theory, molecular geometry, probabilistic immunology, and structural intuition that perceived an isomorphism between two disciplines before it could be articulated — instantiated all ten models simultaneously, and its most significant applications remain, as the framework predicts they must, unknown unknowns. The Human Genome Project demonstrated the same integration at institutional scale: sequencing model organisms before the human genome, embedding ethical responsibility into the project's design from day one, and arriving at its most consequential discovery — that most of the genome was invisible to the instruments used to read it — as a direct consequence of epistemic humility built into the methodology. The Montreal Protocol extended the framework into governance: the recognition that a molecule benign at human scale was lethal at planetary scale, the treatment of the atmosphere as a self-regulating living body with finite tolerance for disturbance, and the institutionalisation of adaptive scientific review together resulted in what Kofi Annan described as the most successful international agreement in history — an outcome whose cognitive pattern is structurally analogous to Leonardo's integrative practice, even if not causally derived from it.

In the case of industrial manufacturing, aside from the Interface example described earlier, the Toyota Production System integrates the widest range of models across a single coherent institutional practice: the theory-experience dialectic (V) is its epistemological foundation; working with the natural logic of the process rather than against it (VIII) is its operational principle; and the structural intuition that waste, inconsistency, and overburden are symptoms of the same underlying misalignment (VI) unifies what would otherwise appear to be separate improvement tools (Ohno, 1988; Spear and Bowen, 1999). These models are not merely present but causally interdependent — remove any one of them and the system reverts to conventional manufacturing, which is precisely what every organisation that copied TPS's tools without its epistemology discovered.

What these cases share — and what distinguishes them from technically competent achievements that address problems without opening new domains — is a set of characteristics that the Leonardian framework makes intelligible: temporal arcs measured in decades rather than grant cycles; a persistent epistemic humility that built acknowledgement of the unknown into the methodology from the outset; cross-domain structural recognition that identified the governing rule beneath apparently unrelated phenomena; and an integration of ethical responsibility with technical practice that treated the two not as competing demands but as aspects of a single commitment to understanding the systems one inhabits. What the framework contributes is their synthesis into a coherent cognitive architecture — one grounded in Leonardo's practice, demonstrated in contemporary evidence, and available, in principle, to anyone willing to cultivate the integrative intelligence it requires. The task, as Leonardo understood it, is not to accumulate knowledge across domains but to perceive the structure beneath apparent diversity: to find, as he put it, the common trunk from which all the branches grow.

Potential Implications for AI design

The following observations are offered speculatively, as questions that the Leonardian framework raises rather than resolves, intended to suggest directions for future investigation rather than to make claims the current paper is not equipped to substantiate.

The mental models articulated in the preceding sections possess formal properties that render them, in principle, amenable to computational representation. This observation invites a fundamental question: could Leonardo's cognitive patterns—derived and formalised from decades of scholarly engagement with his manuscripts and works—be translated into algorithmic frameworks that augment contemporary complex problem solving and creative practice?

Leonardo's knowledge system, as reconstructed from his notebooks and surviving works, exhibits a radial structure centred on direct observation and embodied experience. A diagrammatic reconstruction of this system reveals knowledge domains radiating from a biographical and experiential core—the human body serving as microcosm through which macrocosmic principles become comprehensible. Astronomy connects to optics through light; anatomy connects to engineering through mechanics; art connects to mathematics through

proportion and geometry. The branches multiply and interconnect, forming not a hierarchical taxonomy but a network of relationships discovered through sustained observation and experimental practice. The notation “(A)” indicates areas where Leonardo conducted sustained experimental investigations; circled annotations denote conceptual links between domains mediated by principles of proportion, mechanics, optics, or analogy.

This structure differs fundamentally from the categorical organisations that characterise modern disciplinary knowledge—and, by extension, the training datasets that inform contemporary AI systems. Where AI tends towards either narrow specialisation or undifferentiated generalisation, Leonardo’s system operated through structural analogy: the same principle recognised in different domains, the same mathematical form appearing at different scales. Formalising this capacity—for cross-domain structural recognition rather than topical categorisation—represents perhaps the most significant and most difficult challenge that Leonardo’s models pose to computational instantiation.

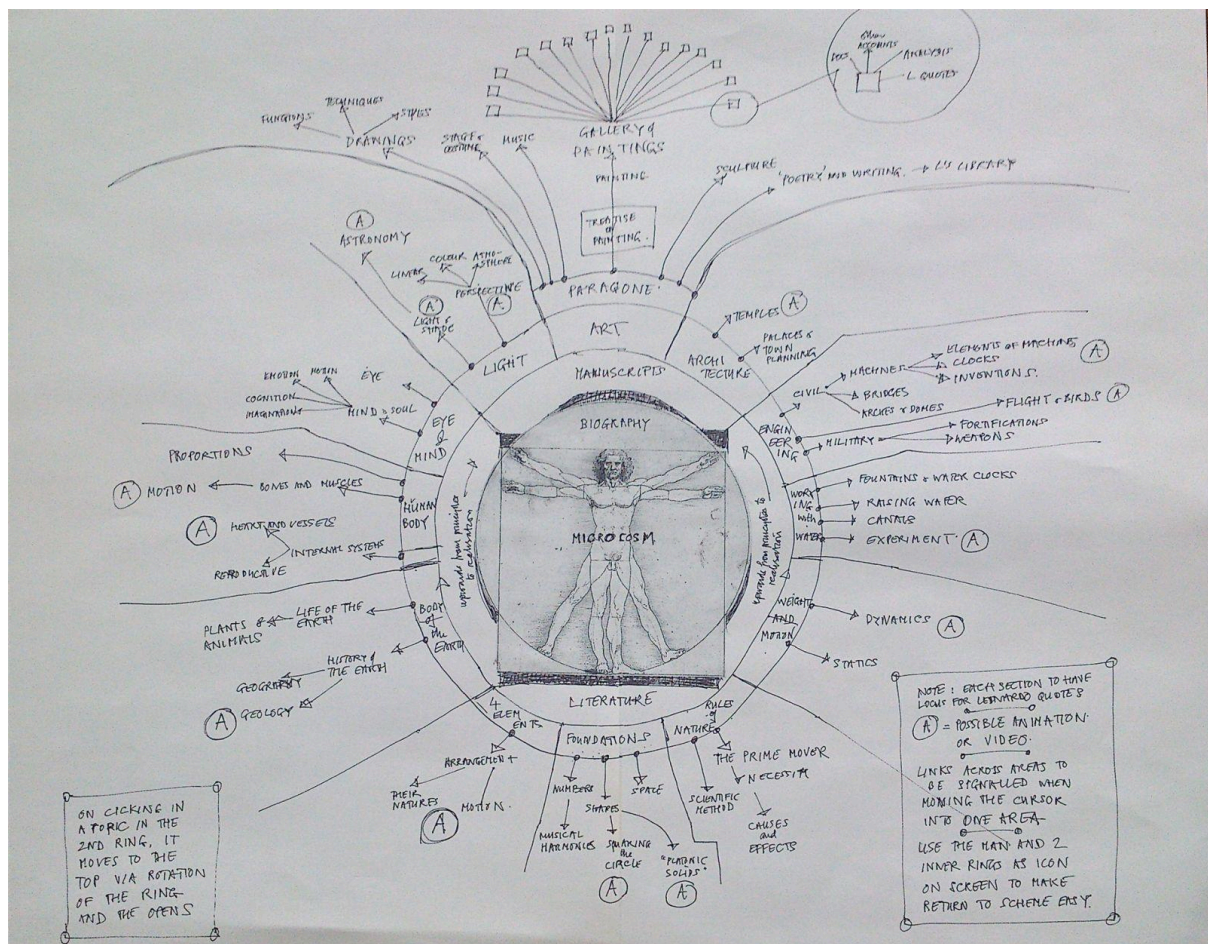


Figure 1. Diagrammatic reconstruction of Leonardo da Vinci’s knowledge system, showing radial integration of diverse domains through shared observational and mathematical principles. Unlike modern disciplinary taxonomies or categorical knowledge graphs, Leonardo’s system centres on

embodied experience (represented by the Vitruvian Man) with domains interconnected through structural relationships rather than topical proximity. (produced by Professor Martin Kemp)

This framework is especially relevant given a specific and measurable limitation of current AI systems: large language models have been observed in several preliminary studies to produce outputs that cluster around statistically central choices in creative and design tasks — a tendency that, if confirmed at scale, would represent a structural bias towards convergence rather than divergence (see e.g. Organisciak and Deyell, 2023, for early empirical work in this direction; the evidence base remains active and not yet definitive).

Conclusion

Leonardo da Vinci's mental models, examined through the evidence assembled in this paper, reveal principles of enduring and, in some respects, increasing relevance. His insistence on the unity of human and natural systems, the primacy of experiential verification alongside theoretical abstraction, the complementary roles of artistic and scientific understanding, and his profound respect for natural force all speak directly to present challenges that resist the application of narrow, single-discipline tools. That these principles are not merely historically interesting but practically generative is demonstrated by the range of contemporary exemplifications gathered here: from the molecular geometry of drug discovery and the probabilistic architecture of modern weather forecasting, to the biomimetic engineering of high-speed rail and the planetary governance of the ozone layer. In each case, the Leonardian mode of thinking — integrative, analogical, humble before the complexity of natural systems — produced outcomes that more specialised approaches had failed to reach.

Perhaps most significantly, Leonardo's embrace of the unknown in his later thought — his recognition that mystery and incomprehension are not failures of inquiry but integral features of the human condition — offers a corrective to the hubristic claims of mastery that characterise much of our scientific and technological culture. The JWST's discovery of galaxies that standard cosmology had not equipped anyone to anticipate, the Human Genome Project's revelation that most of the genome was invisible to the instruments designed to read it, and the mRNA platform's opening of a domain of medicine whose boundaries cannot yet be seen: these are not anomalies in the history of knowledge. They are, as Leonardo's framework predicts, the characteristic signature of genuine encounter with the natural world. His integrative approach, refusing the compartmentalisation of knowledge that characterises modern education and professional practice alike, points not towards a nostalgic ideal but towards a cognitive architecture that remains available — and, the evidence suggests, remains necessary.

The task is not to replicate Leonardo's particular cognitive trajectory, which was shaped by conditions — material, institutional, and biographical — that cannot be reproduced. It is, rather, to recognise in his documented practice a set of cognitive dispositions whose structural properties are identifiable, teachable, and, as the contemporary exemplifications assembled here suggest, generative across domains and eras: rigorous observation held in

tension with theoretical abstraction; aesthetic and analytical perception treated as aspects of a single act; and an honest acknowledgement of the limits of inquiry that is not paralysing but consistently productive. That is what the framework offers — not a model of genius to be admired, but a cognitive architecture to be examined.

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